

Grounded Variance Commentary: A Retrieval-Augmented Assistant

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[Github Link](#)

Abstract

Month-end variance commentary requires accurate figures, policy compliance, and causal support. This study evaluates a local FP&A assistant that separates those authorities: a deterministic engine owns the numbers, retrieval supplies rules and driver evidence, and a local four-billion-parameter model writes only from those inputs. The final system was configured on two monthly closes and evaluated on 20 hold-out cases across eight later closes. The grounded system retrieved the intended driver in 4 of 6 documented cases, achieved 6 of 6 numeric fidelity on documented outputs, and abstained correctly in all 6 evidence-less cases with no over-abstention. Adding the pre-generation gate and numeric verifier improved all-output fidelity to 12 of 12. By comparison, the unfiltered long-context arm achieved 6 of 6 fidelity on documented cases but only 6 of 12 across all generated output. Retrieval remains the main limitation at 67%, and the corpus is small and synthetic. However, when evidence is insufficient, the final system returns work to the analyst rather than inventing a cause.

Keywords: retrieval-augmented generation, abstention, financial planning and analysis, point-in-time retrieval, local language models

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Introduction

Every month-end close, FP&A analysts explain why actual results diverged from budget. The work is repetitive and time-sensitive, so language-model assistance is attractive. However, a fluent paragraph is not useful if the figure is wrong, the policy rule is outdated, or the stated cause is unsupported. Therefore, this project asks a narrower question than whether an LLM can write variance commentary. It asks whether the surrounding system can make the model a writer without making it the authority for the number, rule, or cause. The final design separates those authorities and, when causal evidence is insufficient, abstains and returns the line for analyst follow-up rather than completing the explanation with a plausible guess.

Background

Retrieval-Augmented Generation (RAG) fits this project because the system does not need the model to know more. It needs the model to use the right information at the right time. RAG combines a model's parametric knowledge with retrieved external information, which allows the generation step to be grounded in documents that can be inspected and verified (Lewis et al., 2020). In this project, that matters because the relevant information is not interchangeable. A policy determines whether commentary is required, a driver memo supports the cause, and the financial data provides the figures. Therefore, the design separates those sources rather than giving the model one large body of context and expecting it to determine what should be trusted.

However, retrieval by itself does not solve the problem. Language models can still generate unsupported information even when relevant evidence is available, and a retrieved document is only useful if it is actually the correct document for the question being answered (Ji et al., 2023; Rashkin et al., 2023). This becomes especially important in month-end reporting

because a cause can sound completely reasonable while still being wrong for the period, entity, or account being reviewed. In addition, providing more context is not necessarily safer. Research has shown that models do not use long contexts consistently, particularly when relevant information is buried among other material (Liu et al., 2024). For that reason, this project treats filtering as part of the control structure rather than simply as a way to shorten the prompt.

The finance use case adds another layer. Financial tasks combine numerical reasoning with narrative explanation, and both can fail in different ways. Prior research has shown that numerical reasoning over financial documents remains difficult for language models and that financial question-answering performance can remain weak even when retrieval is available (Chen et al., 2021; Islam et al., 2023). Consequently, this project does not ask the language model to calculate the variance or decide materiality. Those responsibilities remain deterministic. The model is used only where it adds value, which is turning verified figures and retrieved evidence into readable commentary.

Finally, the objective is not simply to produce better-written management commentary. FP&A reporting is expected to connect financial outcomes to business drivers, ownership, and decision-making (Serven & Krumwiede, 2019). At the same time, narrative financial reporting can create the appearance of explanation without necessarily adding reliable information (Merkl-Davies & Brennan, 2007). Therefore, the system is designed around traceability. If evidence supports the cause, the model can draft the explanation. If the evidence is not there, the more appropriate output is to say that the cause is unsupported and return the line to the analyst.

Problem Statement

Useful month-end commentary must be numerically exact, apply the correct version of the reporting policy, and attribute cause only to evidence that existed at the close and covers the

correct entity. Consequently, the design problem is one of authority: numbers belong to deterministic calculation, rules and ownership belong to governed documents, causes belong to driver evidence, and the model should originate none of them.

Moreover, financial information requires a high level of privacy because business and investment decisions can depend on information that has not yet been disclosed publicly. Accuracy and integrity are therefore critical, not only for internal decision-making but also for the stakeholders who may eventually rely on that information. In practice, financial results and their explanations often move through review and approval processes before they are disclosed or published. For that reason, the system must support a controlled workflow in which figures and explanations remain traceable, reviewable, and protected throughout the reporting process.

Data Description

All project data are synthetic; no real organization's policies, employees, customers, or results are used. This allows planted ground truth for the true cause and intended source, consistent with synthetic benchmarking when exact evaluation quantities are difficult to establish in real data (Y. Liu et al., 2021). The fictional manufacturer was calibrated only for plausibility using January 2026 industry-margin data (Damodaran, 2026). The final corpus contains 15 documents and 30 retrievable sections with publication dates, effective windows, versions, scopes, and categories. The full evaluation contains 26 cases across 10 closes; two closes are reserved for configuration and 20 cases across eight closes form the hold-out.

Table 1
Final Data and Evaluation Set

Component	Category	Count	Purpose
Corpus	Rules	3 docs	Policy, KPI definitions, owner register
Corpus	Causes	9 docs	Documented driver evidence
Corpus	Distractors/precedent	3 docs	Wrong-scope, wrong-period, or prior commentary
Evaluation	Immaterial	10 cases	No commentary expected
Evaluation	Evidence-less	8 cases	Tests abstention
Evaluation	Documented	6 cases	Tests retrieval and grounded generation
Evaluation	Version/covenant	2 cases	Tests time/version and policy exceptions

Note. The corpus contains 15 documents and 30 section-level retrieval units. The 26 evaluation cases are distributed across 10 monthly closes.

Methodology

The system separates three authorities. A deterministic Python engine calculates the variance, percentage, price-volume-mix effects when available, and the materiality decision. The model receives those values as verified facts and is instructed not to recompute them. Two materiality rules apply: the dollar threshold and a covenant trigger for movements bringing a lending covenant within 0.25x of its limit.

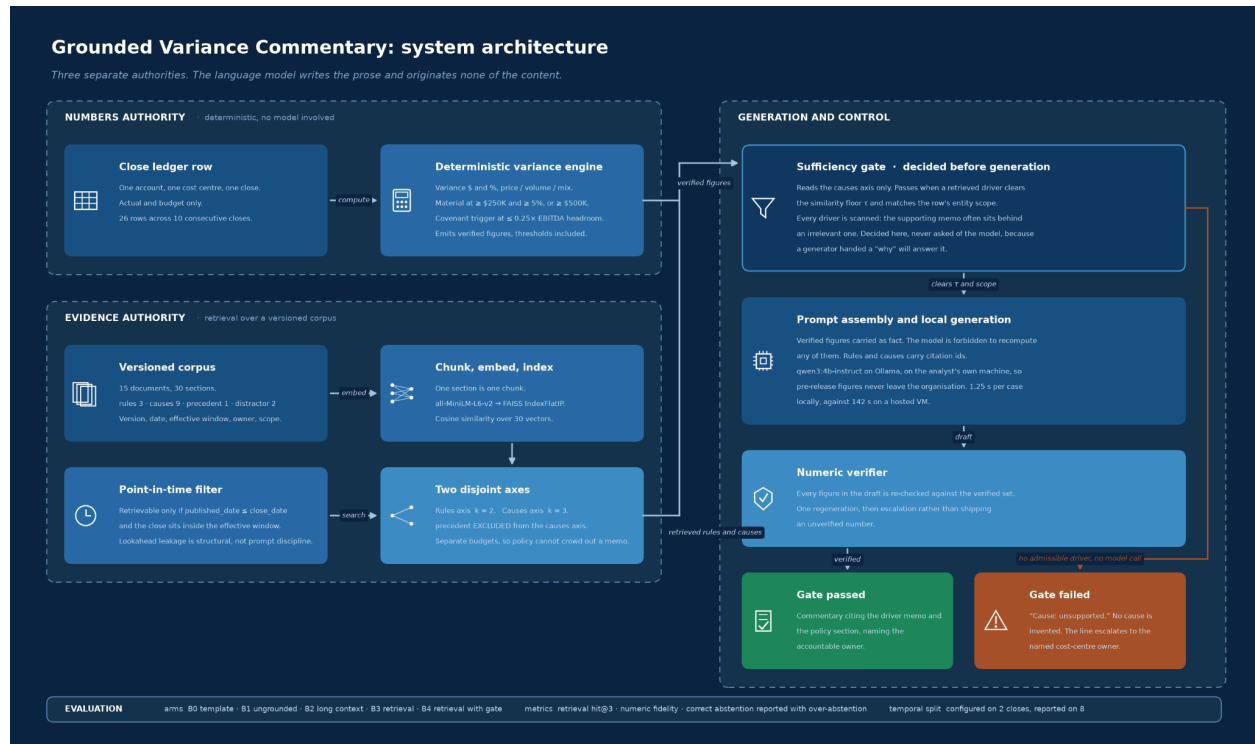
Retrieval operates on two disjoint axes. The rules axis searches policy, definitions, and the owner register; the causes axis searches driver evidence and distractors while excluding prior-period commentary. Before similarity search, a point-in-time mask requires the document to have been published by the close date and to be effective for that period. This control is implemented before generation, so later or expired evidence cannot re-enter through prompt behavior.

Abstention is decided before generation. A sufficiency gate passes only when retrieved driver evidence clears the configured similarity threshold and matches the row's entity scope. If the gate fails, the system returns an unsupported status and escalation rather than asking the model to infer a likely cause. This implements answerability as a system property rather than relying only on cautious prompting (Rajpurkar et al., 2018).

System Architecture

Figure 1 shows the full path for one variance line. The language model is intentionally limited to the generation lane; numbers are computed, while rules and causes are retrieved under explicit controls.

Figure 1
System Architecture Showing the Separation of Numbers, Rules, and Causes



Note. Numeric facts are deterministic. Rules and causal evidence are retrieved under point-in-time and scope controls before local generation.

Implementation

The system was implemented in Python and runs locally on the user's machine. Python manages the variance calculations, document processing, retrieval logic, evaluation functions, and the connection between the model and the user interface. Organizing the project this way keeps the financial calculations and system controls in code, while the language model is used only for the part of the workflow that requires natural-language generation.

For retrieval, the document sections are converted into embeddings using all-MiniLM-L6-v2. These embeddings are stored and searched with FAISS, which allows the system to compare the variance case with the available policy and driver documents and return the most relevant supporting passages. The retrieved content is then passed into the generation step together with the verified financial figures.

Commentary is generated with qwen3:4b-instruct, served locally through Ollama. The model is run at temperature zero with a fixed seed to make the outputs more consistent across repeated runs. Because the model is served locally, the financial figures, retrieved documents, and generated commentary remain on the same machine rather than being sent to an external hosted model.

An additional numeric-verification step checks the figures included in the generated response against the verified inputs and retrieved evidence. This provides a separate control after generation and helps identify cases where the wording may be grounded but a dollar amount or percentage is not supported by the source data.

The system also includes a Streamlit interface so the workflow can be used as a monthly-close process rather than only through code. The interface displays the account, cost center, actual results, budget, variance, and triage status for each line. From there, the user can review the retrieved evidence and generate commentary when appropriate. The same underlying Python pipeline supports both the interface and the evaluation outputs, which helps keep the application behavior and reported results consistent.

Results

Five arms were evaluated on the 20 hold-out cases. B0 is a deterministic template, B1 gives the model verified figures without retrieval, B2 is the unfiltered long-context control, B3 is the proposed two-axis RAG system, and B4 adds the pre-generation gate and numeric verifier.

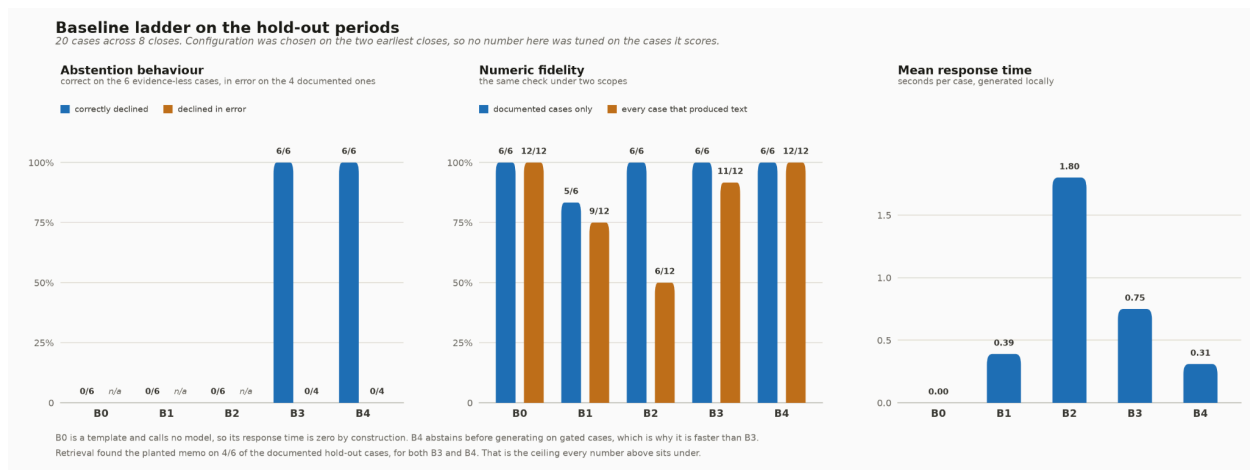
B2 was the most informative control. It achieved 6 of 6 numeric fidelity on documented cases but only 6 of 12 across all generated output because unrelated figures from the larger context appeared in evidence-less cases. The grounded arms retrieved the intended source in 4 of 6 documented cases. B3 achieved 6 of 6 documented-case fidelity and abstained correctly in all 6 evidence-less cases with no over-abstention; B4 improved all-output fidelity from 11 of 12 to 12 of 12.

Table 2
Baseline Ladder on the Hold-Out Periods

Arm	Hit@3	Doc. fidelity	All-output fidelity	Correct abst.	Over-abst.	Mean s
B0	n/a	6/6	12/12	0/6	n/a	0.00
B1	n/a	5/6	9/12	0/6	n/a	0.39
B2	n/a	6/6	6/12	0/6	n/a	1.80
B3	4/6	6/6	11/12	6/6	0/4	0.75
B4	4/6	6/6	12/12	6/6	0/4	0.31

Note. B4 ends gated cases before generation, which contributes to its lower mean response time.

Figure 2
Baseline Ladder Across Abstention, Numeric Fidelity, and Latency



Note. Numeric fidelity is shown at documented-case and all-output scopes. The wider scope exposes unsupported figures on evidence-less cases.

Evaluation

Four measures drive the final evaluation: retrieval hit@3, numeric fidelity, correct abstention, and over-abstention. Response time is also recorded. Correct abstention and over-abstention are always interpreted together because a system that refuses everything would otherwise appear perfect on the first metric alone. This separation follows the broader RAG evaluation principle that retrieval, faithfulness, and answer quality are distinct dimensions (Es et al., 2024).

Configuration was selected using only the two earliest closes, and all headline results come from the eight later closes. One early gate scored perfectly on configuration cases but abstained on every hold-out case, which confirmed why the temporal split was necessary. Across the eight hold-out closes, abstention remained correct, while the two retrieval misses occurred where the intended memo competed with a near-duplicate distractor.

Improvement From the Prototype

The final system improves both the architecture and the evaluation. The earlier prototype used eight hand-picked cases, did not separate configuration from evaluation, measured two evidence-less cases, and did not measure over-abstention. The final version uses 26 cases across 10 closes, a temporal hold-out, five comparison arms, a covenant-sensitive rule, a pre-generation gate, a numeric verifier, and a local Streamlit workflow.

For a like-for-like retrieval comparison, the earlier retrieval configuration was rerun on the same six documented hold-out cases. Hit@3 increased from 2 of 6 to 4 of 6 after prior commentary was removed from the cause-retrieval category. A later top-4 variant reached 5 of 6, but it is not used as the headline result because it was tested after a failed hold-out case had already been inspected.

Table 3*Prototype and Final System Comparison*

Area	Prototype	Final
Retrieval, same hold-out cases	2/6 (33%)	4/6 (67%)
Numeric fidelity, documented	4/5	6/6
Correct abstention	2/2	6/6
Over-abstention	Not measured	0/4
Test design	8 hand-picked	20 hold-out across 8 closes
Comparison arms	1	5
Materiality	Dollar only	Dollar + covenant
Serving	Hosted VM	Local + Streamlit

Note. The retrieval comparison is like-for-like. Other rows summarize the broader prototype versus final evaluation and therefore use different sample sizes.

Discussion

The main result is not the 67% retrieval rate but what happens when retrieval misses. Both documented misses became abstentions rather than wrong causes, turning an invisible reporting error into visible unfinished work. The B2 comparison also shows that filtering performs control work beyond reducing context size because publication date, effective window, category, and entity scope determine whether evidence should be available at all.

The trade-off is lower automation when evidence is weak in exchange for a safer failure mode. Some lines still return to the analyst, but the system does not complete the record with a cause it cannot support. The wider all-output fidelity measure reinforced the same lesson by exposing unsupported figures that a documented-only metric would have hidden.

Limitations and Risks

The corpus is synthetic and small: 15 documents are enough to create meaningful distractors but not enough to estimate performance on a repository containing hundreds or thousands of finance documents. The hold-out also contains only six documented and six evidence-less material cases, so the results establish direction rather than statistical confidence.

Retrieval remains the binding constraint at 4 of 6 documented cases. Cross-encoder reranking (Nogueira & Cho, 2019) and alternative retrieval granularity (T. Chen et al., 2024) are the most direct next experiments. Results also come from one generation model, one embedding

model, and one seed, and planted ground truth does not guarantee that a finance reviewer would consider the authored gold document the only acceptable evidence.

One B4 defect must be corrected before further use. On gated refusals, the escalation line currently takes the first retrieved owner-register section without confirming that it covers the row's cost center. Two of five escalation lines in the reported run therefore cite the wrong register section. The abstention decision remains correct and the headline metrics are unaffected, but the citation itself is unsupported.

Responsible AI and Ethics

Human review is part of the intended workflow because the system cannot prove that a retrieved memo is the correct business explanation or detect that a required memo was never written. Local inference keeps the demonstrated financial figures on the machine. The controls map to the NIST AI Risk Management Framework (Tabassi, 2023): human review supports GOVERN, data boundaries support MAP, hold-out evaluation supports MEASURE, and abstention with escalation supports MANAGE. A future real-document pilot should add independent human review and broader RAG evaluation, including reference-free measures such as those proposed by Es et al. (2024).

Conclusion

The final system separates numeric, policy, and causal authority, filters evidence by time and scope, and decides whether evidence is sufficient before generation. It retrieved the intended source in 4 of 6 documented hold-out cases, maintained complete documented-case numeric fidelity, abstained correctly in all 6 evidence-less cases with no over-abstention, and reached 12 of 12 all-output fidelity with the verifier. It is not production-ready: retrieval remains the primary limitation, the dataset is synthetic, and the owner-register citation defect must be fixed. However,

when evidence is insufficient, the system returns the line to the analyst rather than inventing a cause, which is the improvement that matters most.

References

- Chen, T., Wang, H., Chen, S., Yu, W., Ma, K., Zhao, X., Zhang, H., & Yu, D. (2024). Dense X retrieval: What retrieval granularity should we use? In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (pp. 15159-15177). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.emnlp-main.845>
- Chen, Z., Chen, W., Smiley, C., Shah, S., Borova, I., Langdon, D., Moussa, R., Beane, M., Huang, T.-H., Routledge, B., & Wang, W. Y. (2021). FinQA: A dataset of numerical reasoning over financial data. In M.-F. Moens, X. Huang, L. Specia, & S. W. Yih (Eds.), *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 3697-3711). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2021.emnlp-main.300>
- Damodaran, A. (2026, January). Operating and net margins by industry [Data set]. New York University Stern School of Business. Retrieved August 30, 2026, from https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/margin.html
- Es, S., James, J., Espinosa Anke, L., & Schockaert, S. (2024). RAGAs: Automated evaluation of retrieval augmented generation. In N. Aletras & O. De Clercq (Eds.), *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations* (pp. 150-158). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.eacl-demo.16>
- Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, M., & Wang, H. (2023). Retrieval-augmented generation for large language models: A survey (arXiv:2312.10997). arXiv. <https://doi.org/10.48550/arXiv.2312.10997>

- Islam, P., Kannappan, A., Kiela, D., Qian, R., Scherrer, N., & Vidgen, B. (2023). FinanceBench: A new benchmark for financial question answering (arXiv:2311.11944). arXiv.
<https://doi.org/10.48550/arXiv.2311.11944>
- Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), Article 248. <https://doi.org/10.1145/3571730>
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. In H. Larochelle, M. Ranzato, R. Hadsell, M. F. Balcan, & H. Lin (Eds.), *Advances in Neural Information Processing Systems* (Vol. 33, pp. 9459-9474). Curran Associates.
- Li, Z., Li, C., Zhang, M., Mei, Q., & Bendersky, M. (2024). Retrieval augmented generation or long-context LLMs? A comprehensive study and hybrid approach. In F. Dernoncourt, D. Preotiu-Pietro, & A. Shimorina (Eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track* (pp. 881-893). Association for Computational Linguistics.
<https://doi.org/10.18653/v1/2024.emnlp-industry.66>
- Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12, 157-173. https://doi.org/10.1162/tacl_a_00638
- Liu, Y., Khandagale, S., White, C., & Neiswanger, W. (2021). Synthetic benchmarks for scientific research in explainable machine learning (arXiv:2106.12543). arXiv.
<https://doi.org/10.48550/arXiv.2106.12543>

- Merkel-Davies, D. M., & Brennan, N. M. (2007). Discretionary disclosure strategies in corporate narratives: Incremental information or impression management? *Journal of Accounting Literature*, 26, 116-196.
- Nogueira, R., & Cho, K. (2019). Passage re-ranking with BERT (arXiv:1901.04085). arXiv. <https://doi.org/10.48550/arXiv.1901.04085>
- Rajpurkar, P., Jia, R., & Liang, P. (2018). Know what you don't know: Unanswerable questions for SQuAD. In I. Gurevych & Y. Miyao (Eds.), *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)* (pp. 784-789). Association for Computational Linguistics. <https://doi.org/10.18653/v1/P18-2124>
- Rashkin, H., Nikolaev, V., Lamm, M., Aroyo, L., Collins, M., Das, D., Petrov, S., Tomar, G. S., Turc, I., & Reitter, D. (2023). Measuring attribution in natural language generation models. *Computational Linguistics*, 49(4), 777-840. https://doi.org/10.1162/coli_a_00486
- Serven, L., & Krumwiede, K. (2019, July 2). Key principles of effective financial planning and analysis [Statement on Management Accounting]. Institute of Management Accountants. <https://www.imanet.org/research-publications/statements-on-management-accounting/key-principles-of-effective-financial-planning-and-analysis>
- Tabassi, E. (2023). Artificial intelligence risk management framework (AI RMF 1.0) (NIST AI 100-1). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.AI.100-1>

AI Usage Disclosure

AI Tools Used

Claude was used throughout design, drafting, code generation, and verification under my direction and review. Claude Code was used for the local Python package, experiment scripts, Streamlit interface, notebook, and figures. Ollama with qwen3:4b-instruct is the local model runtime under evaluation rather than a drafting aid, while sentence-transformers and FAISS are the retrieval stack under test.

How I Used AI

Module 8 moved the project from Google Colab to a local machine. Claude helped restructure the Module 7 notebook into the src/gvc Python package, implement the five comparison arms B0 to B4, extend the evaluation from eight hand-picked probes to 26 cases across 10 consecutive closes, add the covenant materiality rule and three new driver memos, build the Streamlit interface, generate the architecture and results figures, and draft repository materials from executed output files.

Prompts Used

Important prompt categories included porting the Colab notebook into a package that first reproduces the Module 7 result; designing an evaluation around consecutive monthly closes; building the baseline ladder; reviewing metric definitions for scopes that could flatter the system; and drafting the README and notebook narrative from executed result files while checking alignment and inconsistencies.

What I Verified Myself

I requested the temporal split after the earlier design allowed configuration and scoring on the same cases. I ran tests/test_parity.py before trusting new measurements, which caught a pandas date-parsing difference that would have silently broken the point-in-time filter. I also

expanded abstention into the paired correct-abstention and over-abstention measures, widened numeric fidelity from documented cases to all generated output, and rechecked the gate after discovering that the correct memo could appear at rank 2 or 3 rather than rank 1.

Every number in the README, notebook, and article traces to a result file written by the build or test scripts. I ran the full pipeline on my own machine, executed the notebook top to bottom, and tested the Streamlit interface on both a documented case and an evidence-less case to confirm that the abstention state renders correctly.

Failures or Limitations

Claude's first sufficiency gate carried forward a 0.45 absolute similarity threshold from Module 7. On the final corpus, similarity scores cluster between approximately 0.34 and 0.45, so the gate abstained on every hold-out case. The failure became visible because the mean response time fell to 0.00 seconds, which was not physically plausible for a generation arm. The final corpus also remained smaller than the 38-document proposal target so the project would stay light enough to replicate locally.

Academic Integrity Statement

I confirm that AI was used as a learning, development, and drafting support tool under my direction. I made the design decisions, ran every experiment on my own machine, verified each reported result against the output files produced by those runs, and take responsibility for the final submitted work.

Addendum: UI Screenshots

Grounded Variance Commentary

An FP&A month-end assistant that cites its evidence, or declines to answer.

Sign in

Your name

L. Santos

Password

not checked, leave it blank

Role

☒ Reviewer

Everything an analyst sees, plus the method view: the five comparison approaches, the raw similarity scores, the planted-source marker, and the assembled prompts.

☐ Analyst

The close review, the cross-period comparison, and the measured evaluation scores. Only the method internals are hidden, which is how the tool would look to a finance user.

Sign in

This prototype does not authenticate anyone. The password box is not checked against anything, no password is stored or transmitted, and there is no account to create. Your name is held in memory for this session alone, so that drafted commentary can be signed the way a real close would be. Choosing a role changes what the interface shows you and nothing else. Everything runs on this machine.

This system is a classroom prototype. Outputs should be reviewed by a human before real-world use.

Grounded Variance Commentary

A month-end assistant for variance commentary. Every figure is computed rather than written by a model, every cause is attributed to a source document, and the system says so plainly when it cannot explain a line.

This system is a classroom prototype. Outputs should be reviewed by a human before real-world use.

Open a close

Opening a close fetches its lines and applies the reporting rules. Both are arithmetic. Searching for evidence and writing the commentary are separate steps that you trigger yourself.

Close	Lines	Movement under review	Need commentary	
June 2023		3	1,141,000	2
November 2023		3	1,108,000	2
February 2024		3	1,649,000	2
May 2024		3	666,000	2
September 2024		3	2,750,000	2
December 2024		2	414,000	1
March 2025		2	106,000	1
June 2025		2	686,000	1
August 2025		2	627,000	1
October 2025		3	2,321,000	2

Close period

Jun 2023

Nov 2023

Feb 2024

May 2024

Sep 2024

Dec 2024

Mar 2025

Jun 2025

Aug 2025

Oct 2025

→ Open the October 2025 close

26 variance lines across 10 closes, 16 of which need commentary under the reporting policy. Pick a close below and open it. You can move between closes at any time without losing what you have already run.

Or load a close from a file

The ten closes above are fixed in the code, which is practical for an evaluation and not for anything else. This is how a month actually arrives: a general-ledger export of actuals against budget, which the same deterministic engine runs over unchanged.

Column	Required	What it holds
period	yes	the close, as YYYY-MM
account	yes	code and name, for example 5000 - Cost of Goods Sold
cost_centre	yes	CC-2100, CC-2200, CC-3100, CC-5100 or CC-6100
region	yes	Group, North America or EMEA
product_line	yes	All, or a named line
actual	yes	the actual, with or without \$ and separators
budget	yes	the budget
price_effect, volume_effect, mix_effect	no	flexible-budget split, revenue lines only
covenant_headroom, covenant_limit	no	turns of EBITDA. Headroom at or below 0.25 requires commentary whatever the dollar amount

There is no column for the cause, the supporting document, or the answer, because a ledger export carries none of those. That is the point: a close read from a file is **reviewed, never scored**, and the app marks it that way everywhere it appears.

Any close loaded here is kept apart from the ten evaluated closes and is excluded from the cross-period comparison.

[Download the format as a CSV](#)

Contains the October 2025 close as a worked example of the shape.

Upload a close

Upload

200MB per file • CSV

How this works

1. **Reporting check.** Decides whether a line is large enough to need commentary at all. Pure arithmetic against the policy thresholds. Runs when you open a close.
2. **Evidence search.** Looks for a document that could support a cause for a flagged line. A similarity search over 30 document sections, with no language model involved. Runs when you ask.
3. **Commentary.** Writes the wording, or refuses to state a cause and escalates to the cost-centre owner. The only step that calls a language model, and you set the rules it has to obey. Runs when you ask.

Where this data comes from

These ten closes are fixed in the code and are the evaluation set behind every number in the accompanying article. They are a representative extract, not a live feed: a real month-end file would hold hundreds of lines, most of them below the threshold.

Opening a close is an exact lookup by period, so a figure is never approximated. That is deliberate, and it is the opposite of how the document search works. Searching for a cause is a similarity search, because nobody knows in advance which memo explains a movement, whereas a figure that is merely similar is simply wrong.

A new month arrives as a general-ledger export of actuals against budget, which is what the loader above accepts. The same deterministic engine runs over it unchanged, so a close read from a file goes through exactly the same rules as one shipped in the code.

The document set is fixed at 15 documents, so a genuinely new month has few driver memos that cover it and will correctly refuse to explain most of its lines. That is the system working, not failing.

June 2023 close

← All closes

This system is a classroom prototype. Outputs should be reviewed by a human before real-world use.

Close review All periods Results Method

Lines in this close

3

2 need commentary

Movement under review

\$1,141,000

sum of the absolute variances

Material movement

\$1,130,000

on lines that need commentary

Unexplained

not searched yet

run the evidence search

Actuals and budgets are not added across revenue and expense accounts, because that total would not mean anything. Absolute variance is the meaningful aggregate: it is the dollar movement sitting in front of you.

Where this close has got to

Stage 1 Reporting check

Done. 2 of 3 lines need commentary.

Stage 2 Evidence search

Not run.

Search 2 flagged lines

Stage 3 Commentary

Waiting for the evidence search.

How should it answer? - the rules the article measures

Every line in this close

Status	Account	Cost centre	Actual \$	Budget \$	Variance \$	Variance %	Evidence	Commentary	Owner
☐ Needs evidence	6100 - Legal & Professional	CC-6100	1,268,000	633,000	635,000	+100.3%	Not searched yet	Not drafted yet	M. Duarte
☐ Needs evidence	5000 - Cost of Goods Sold	CC-3100	5,495,000	5,000,000	495,000	+9.9%	Not searched yet	Not drafted yet	T. Bianchi
☐ Below threshold	6200 - Marketing Operating Expense	CC-2200	386,000	375,000	11,000	+2.9%	Not required	Not required	P. Nunes

2 awaiting the evidence search 1 below threshold

Open a line

Legal & Professional Cost of Goods Sold Marketing Operating Expense

June 2023 close

← All closes

This system is a classroom prototype. Outputs should be reviewed by a human before real-world use.

Close review

All periods

Results

Method

Lines in this close

3

2 need commentary

Movement under review

\$1,141,000

sum of the absolute variances

Material movement

\$1,130,000

on lines that need commentary

Unexplained

\$635,000

no document supports a cause

Actuals and budgets are not added across revenue and expense accounts, because that total would not mean anything. Absolute variance is the meaningful aggregate: it is the dollar movement sitting in front of you.

Where this close has got to

Stage 1 Reporting check

Done. 2 of 3 lines need commentary.

Stage 2 Evidence search

Done. 1 of 2 flagged lines have a supporting document.

🔍 Search again

Stage 3 Commentary

Not run. 2 lines are waiting.

> How should it answer? - the rules the article measures

🔗 Draft commentary for 2 lines

Runs the whole close under the rules above. A line with no supporting evidence is refused without calling the model, so it costs no time.

Every line in this close

Status	Account	Cost centre	Actual \$	Budget \$	Variance \$	Variance %	Evidence	Commentary	Owner
☐ Chase the owner	6100 - Legal & Professional	CC-6100	1,268,000	633,000	635,000	+100.3%	Nothing close enough	Not drafted yet	M. Duarte
☐ Explanation found	5000 - Cost of Goods Sold	CC-3100	5,495,000	5,000,000	495,000	+9.9%	Document found	Not drafted yet	T. Bianchi
☐ Below threshold	6200 - Marketing Operating Expense	CC-2200	386,000	375,000	11,000	+2.9%	Not required	Not required	P. Nunes

1 to chase

1 ready to explain

1 below threshold

Open a line

Legal & Professional

Cost of Goods Sold

Marketing Operating Expense

6100 - Legal & Professional

CC-6100 - Group - All - June 2023 close - owner M. Duarte, Group Financial Controller

The figures

Actual

\$1,268,000

Budget

\$633,000

Variance \$

+\$635,000

Variance %

+100.3%

Computed by a deterministic calculation, not by a model. The model receives these as fact and is forbidden to recompute any of them.

> Every figure handed to the model

Does this line need commentary?

Yes. The variance is \$635,000 (+100.3%). Over \$500,000, so commentary is required at account and cost-centre level.

What the document search found

No document in the set is close enough to this line to support a cause. The search returned results, however none of them are about this account.

Therefore no cause can be attributed. This line has to go back to M. Duarte, Group Financial Controller.

Documents that might explain the movement

Document	Type	Section	Covers	Published	Match strength
Steel Index Movement	Memo	\$2 Index Movement and Cost Impact	Drive Assemblies	2023-06-12	0.18
Resin Cost Movement	Memo	\$1 Summary	Precision Fittings	2023-06-14	0.16

Policy and ownership

Document	Type	Section	Covers	Published	Match strength
Variance Commentary Policy	Policy	\$4.3 Covenant-Sensitive Disclosure	All	2022-12-15	0.46
Cost Centre Owner Register	Owner register	Register – General and Administrative	All	2023-01-03	0.32

Only documents that existed at the June 2023 close were searched. A memo written afterwards is invisible to this line, which is what stops a later explanation being applied to an earlier close.

> Read the retrieved passages in full

Ask for the commentary

Nothing has been asked of the model for this line yet. Read the request below, change the rules above if you want to, then send it. Drafting takes

Cost Centre Owner Register

Owner register

Register — General and Administrative

All

2023-01-03

0.32

Only documents that existed at the June 2023 close were searched. A memo written afterwards is invisible to this line, which is what stops a later explanation being applied to an earlier close.

Read the retrieved passages in full

Procurement Memo — Steel Index Movement - \$2 Index Movement and Cost Impact - published 2023-06-12 - covers Drive Assemblies

The 18% increase in the steel index feeds directly into Drive Assemblies unit cost, which carries the highest steel content of our product lines. The estimated effect on June cost of goods sold for Drive Assemblies is an increase of approximately USD 495,000 versus the frozen budget. Precision Fittings and Aftermarket Parts are minimally affected.

Procurement Memo — Resin Cost Movement - \$1 Summary - published 2023-06-14 - covers Precision Fittings

To: Operations Controller. From: Procurement. Date: 14 June 2023. Subject: Resin cost movement and impact on Precision Fittings. The engineered-resin index rose 4% in June, affecting Precision Fittings, which carries the highest resin content. The estimated June cost impact is modest and below the commentary threshold.

Variance Commentary Policy - \$4.3 Covenant-Sensitive Disclosure - published 2022-12-15 - covers All

Irrespective of the dollar thresholds in §4.2, commentary is required for any movement that brings a lending-covenant metric within 0.25x of its limit. Covenant matters are cross-referenced to the Covenant Compliance and Disclosure Policy.

Cost Centre Owner Register - Register — General and Administrative - published 2023-01-03 - covers All

CC-6100 General and Administrative is owned by M. Duarte, Group Financial Controller; escalation to the CFO. Legal and professional accruals within CC-6100 are reviewed by the Group Financial Controller.

Ask for the commentary

Cause: unsupported

The system declined to state a cause because no document supports one. It escalated to **M. Duarte, Group Financial Controller**. This is the intended behaviour, not a failure to answer.

6100 - Legal & Professional variance \$635,000 (+100.3%) in 2023-06 [1]. Cause: unsupported, no admissible driver evidence retrieved (no driver clears tau 0.25 (best 0.182)). Escalate to the cost-centre owner. Owner per D04 Register — General and Administrative.

Figures check

All traced

every number matches a source

Cause

Not stated

escalated to the owner

Time taken

0.00s

no model was called

Prepared by Unnamed preparer on 30 August 2026. Produced with model with searched evidence and an evidence check.

The time is zero because the system refused on the evidence alone and never called the model.

Ask again

5000 · Cost of Goods Sold

CC-3100 · North America · Drive Assemblies · June 2023 close · owner T. Bianchi, Operations Controller

The figures

Actual

\$5,495,000

Budget

\$5,000,000

Variance \$

+\$495,000

Variance %

+9.9%

Computed by a deterministic calculation, not by a model. The model receives these as fact and is forbidden to recompute any of them.

Every figure handed to the model

Does this line need commentary?

Yes. The variance is \$495,000 (+9.9%). Over \$250,000 and over 5%, so commentary is required at account and cost-centre level.

What the document search found

A supporting document was found: **Procurement Memo — Steel Index Movement** (D11). It covers this line's part of the business and it existed at the June 2023 close, so a cause drafted from it can be attributed to a source.

Documents that might explain the movement

Document	Type	Section	Covers	Published	Match strength
Steel Index Movement	Memo	\$2 Index Movement and Cost Impact	Drive Assemblies	2023-06-12	0.51
Resin Cost Movement	Memo	\$1 Summary	Precision Fittings	2023-06-14	0.35

Policy and ownership

Document	Type	Section	Covers	Published	Match strength
Variance Commentary Policy	Policy	\$4.2 Materiality Thresholds	All	2022-12-15	0.46
Cost Centre Owner Register	Owner register	Register — Sales and Commercial	All	2023-01-03	0.39

Only documents that existed at the June 2023 close were searched. A memo written afterwards is invisible to this line, which is what stops a later explanation being applied to an earlier close.

Read the retrieved passages in full

Ask for the commentary

Cause stated and attributed

Drafted from the retrieved evidence above, and every figure was re-checked against the verified calculation before this was shown.

5000 · Cost of Goods Sold

CC-3100 · North America · Drive Assemblies · June 2023 close · owner T. Bianchi, Operations Controller

The figures

Actual \$5,495,000	Budget \$5,000,000	Variance \$ +\$495,000	Variance % +9.9%
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Computed by a deterministic calculation, not by a model. The model receives these as fact and is forbidden to recompute any of them.

> Every figure handed to the model

Does this line need commentary?

Yes. The variance is \$495,000 (+9.9%). Over \$250,000 and over 5%, so commentary is required at account and cost-centre level.

What the document search found

A supporting document was found: **Procurement Memo — Steel Index Movement** (D11). It covers this line's part of the business and it existed at the June 2023 close, so a cause drafted from it can be attributed to a source.

Documents that might explain the movement

Document	Type	Section	Covers	Published	Match strength
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Resin Cost Movement	Memo	\$1 Summary	Precision Fittings	2023-06-14	0.35

Policy and ownership

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Variance Commentary Policy	Policy	\$4.2 Materiality Thresholds	All	2022-12-15	0.46
Cost Centre Owner Register	Owner register	Register — Sales and Commercial	All	2023-01-03	0.39

Only documents that existed at the June 2023 close were searched. A memo written afterwards is invisible to this line, which is what stops a later explanation being applied to an earlier close.

> Read the retrieved passages in full

Ask for the commentary

Cause stated and attributed
Drafted from the retrieved evidence above, and every figure was re-checked against the verified calculation before this was shown.

Stage 3 · Commentary

Done. Every line that needs commentary has been drafted.

> How should it answer? - the rules the article measures

Every line in this close

Status	Account	Cost centre	Actual \$	Budget \$	Variance \$	Variance %	Evidence	Commentary	Owner
☐ Chase the owner	6100 · Legal & Professional	CC-6100	1,268,000	633,000	635,000	+100.3%	Nothing close enough	Declined	M. Duarte
☐ Explanation found	5000 · Cost of Goods Sold	CC-3100	5,495,000	5,000,000	495,000	+9.9%	Document found	Drafted	T. Bianchi
☐ Below threshold	6200 · Marketing Operating Expense	CC-2200	386,000	375,000	11,000	+2.9%	Not required	Not required	P. Nunes

1 to chase 1 ready to explain 1 below threshold

Open a line

Legal & Professional Cost of Goods Sold **Marketing Operating Expense**

6200 · Marketing Operating Expense

CC-2200 · Group · All · June 2023 close · owner P. Nunes, Head of Marketing

The figures

Actual \$386,000	Budget \$375,000	Variance \$ +\$11,000	Variance % +2.9%
---------------------	---------------------	--------------------------	---------------------

Computed by a deterministic calculation, not by a model. The model receives these as fact and is forbidden to recompute any of them.

> Every figure handed to the model

Does this line need commentary?

No. The variance is \$11,000 (+2.9%), which is under both reporting limits: \$500,000 on its own, or \$250,000 together with 5% of budget (\$18,750 for this line). Policy prohibits commentary on an immaterial variance, so the document search will not run on it and no commentary will be written. The line was reviewed and excluded, which is a reportable answer in its own right.

June 2023 close

← All closes

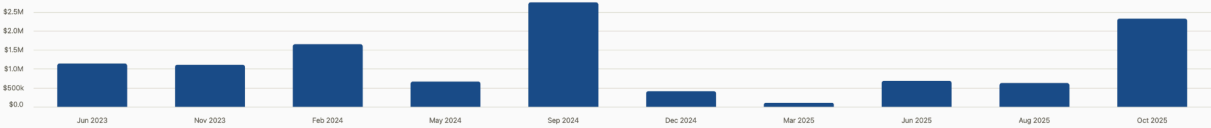
This system is a classroom prototype. Outputs should be reviewed by a human before real-world use.

Close review All periods Results

Every close in the evaluation set

The same review across all ten closes. Replication month over month is one of the project's findings, and it is only visible when the periods are put side by side.

Dollar movement under review, close by close



Close	Lines	Need commentary	Searched	Explanation found	To chase	Below threshold
June 2023	3	2	yes		1	1
November 2023	3	2	no		None	None
February 2024	3	2	no		None	None
May 2024	3	2	no		None	None
September 2024	3	2	no		None	None
December 2024	2	1	no		None	None
March 2025	2	1	no		None	None
June 2025	2	1	no		None	None
August 2025	2	1	no		None	None
October 2025	3	2	no		None	None

16 of 26 lines need commentary. 10 are below the reporting threshold and receive no commentary by design.

The evidence search has been run on 1 of 10 closes.

Search every close

Actual against budget, by account, across all ten closes

Account	Lines	Actual \$	Budget \$	Variance \$	Variance %	Need commentary
4000 - Revenue	4	25,497,000	25,125,000	372,000	+1.5%	4
5000 - Cost of Goods Sold	4	35,485,000	33,125,000	2,360,000	+7.1%	4

Actual against budget, by account, across all ten closes

Account	Lines	Actual \$	Budget \$	Variance \$	Variance %	Need commentary
4000 - Revenue	4	25,497,000	25,125,000	372,000	+1.5%	4
5000 - Cost of Goods Sold	4	35,485,000	33,125,000	2,360,000	+7.1%	4
6100 - Legal & Professional	4	6,455,000	2,532,000	3,923,000	+154.9%	4
6200 - Marketing Operating Expense	8	3,048,000	3,000,000	48,000	+1.6%	0
6300 - Distribution Expense	2	2,796,000	2,100,000	696,000	+33.1%	2
6500 - R&D Operating Expense	3	2,328,000	2,025,000	301,000	+14.9%	1
7000 - Interest Expense	1	1,146,000	1,042,000	104,000	+10.0%	1

Totals within an account are meaningful because actual and budget are measured the same way. They are not added across accounts, because a revenue dollar and a cost dollar do not sum to anything. Marketing carries eight of the ten lines that never breach the threshold, which is why the routine half of this ledger looks repetitive.

Every line

Status	Close	Account	Cost centre	Actual \$	Budget \$	Variance \$	Variance %	Evidence	Commentary
Below threshold	Feb 2024	6200 - Marketing Operating Expense	CC-2200	371,000	375,000	(4,000)	-1.1%	Not required	Not required
Needs evidence	May 2024	6500 - R&D Operating Expense	CC-5100	956,000	675,000	281,000	+41.6%	Not searched yet	Not drafted yet
Needs evidence	May 2024	6100 - Legal & Professional	CC-6100	1,004,000	633,000	371,000	+58.6%	Not searched yet	Not drafted yet
Below threshold	May 2024	6200 - Marketing Operating Expense	CC-2200	389,000	375,000	14,000	+3.7%	Not required	Not required
Needs evidence	Sep 2024	6100 - Legal & Professional	CC-6100	3,033,000	633,000	2,400,000	+379.1%	Not searched yet	Not drafted yet
Needs evidence	Sep 2024	6300 - Distribution Expense	CC-3100	1,395,000	1,050,000	345,000	+32.9%	Not searched yet	Not drafted yet
Below threshold	Sep 2024	6200 - Marketing Operating Expense	CC-2200	380,000	375,000	5,000	+1.3%	Not required	Not required
Needs evidence	Dec 2024	4000 - Revenue	CC-2100	4,095,000	4,500,000	(405,000)	-9.0%	Not searched yet	Not drafted yet
Below threshold	Dec 2024	6200 - Marketing Operating Expense	CC-2200	384,000	375,000	9,000	+2.4%	Not required	Not required

These 26 lines are the whole evaluation set. A real month-end file would hold hundreds of lines with most of them immaterial, and would show far more variety in the routine half of the ledger than ten lines can. Nothing here is padded or illustrative: every line drives a result reported in the article, so adding rows to make a close look busier would break the correspondence between the interface and the measured results.

June 2023 close

← All closes

This system is a classroom prototype. Outputs should be reviewed by a human before real-world use.

Close review All periods Results

Evaluation results

Every number on this page is read from a file in results/ when the page loads. Nothing here is transcribed, so the interface cannot drift from the evaluation. Each table names the file it came from and the day that file was written.

Headline, hold-out closes only

Arm	Approach	Cases	Right memo found	Figures traced, documented	Figures traced, all output	Correctly declined	Declined in error	Mean seconds
B0	Fixed template, no model	20	n/a	6/6	12/12	0/6	n/a	0.00
B1	Model with no evidence	20	n/a	5/6	9/12	0/6	n/a	0.39
B2	Model with the whole document set	20	n/a	6/6	6/12	0/6	n/a	1.80
B3	Model with searched evidence	20	4/6	6/6	11/12	6/6	0/4	0.75
B4	Model with searched evidence and an evidence chec	20	4/6	6/6	12/12	6/6	0/4	0.31

Source results/benchmark_results.csv , written 28 August 2026. Configuration was chosen on the two earliest closes and every figure above comes from the eight later ones.

The two findings this rests on

Retrieval failure degrades into abstention, not a wrong cause.

The search misses on two of the six documented hold-out cases. On both, the system refuses to state a cause and escalates to the named cost-centre owner instead of inventing one. A miss costs a chase, not a false explanation in the board pack.

A metric's scope can hide the failure it exists to catch.

Numeric fidelity scored only on documented cases cannot see a figure invented on a case that abstained. Widening it to every case that produced text drops the long-context arm from 6/6 to 6/12 and reverses the ranking of the last two arms. Both scopes are reported everywhere.

Replication across the reported closes

Close	Cases	Right memo found	Correctly declined	Figures traced
2024-02	3	1/1	1/1	3/3
2024-05	3	1/1	1/1	3/3
2024-09	3	n/a	2/2	2/3
2024-12	2	1/1	n/a	2/2
2025-03	2	1/1	n/a	2/2

One close is not a result. The same behaviour holding across eight consecutive closes, against a document set that grows over time, is what makes it one.

n/a means the metric is undefined for that close, not that it failed. A close with no documented case has no retrieval to score, and a close with no evidence-less case has no abstention to score.

Replication across the reported closes

Close	Cases	Right memo found	Correctly declined	Figures traced
2024-02	3	1/1	1/1	3/3
2024-05	3	1/1	1/1	3/3
2024-09	3	n/a	2/2	2/3
2024-12	2	1/1	n/a	2/2
2025-03	2	1/1	n/a	2/2
2025-06	2	n/a	1/1	2/2
2025-08	2	0/1	n/a	2/2
2025-10	3	0/1	1/1	3/3

One close is not a result. The same behaviour holding across eight consecutive closes, against a document set that grows over time, is what makes it one.

n/a means the metric is undefined for that close, not that it failed. A close with no documented case has no retrieval to score, and a close with no evidence-less case has no abstention to score.

Source results/per_period_results.csv , written 28 August 2026.

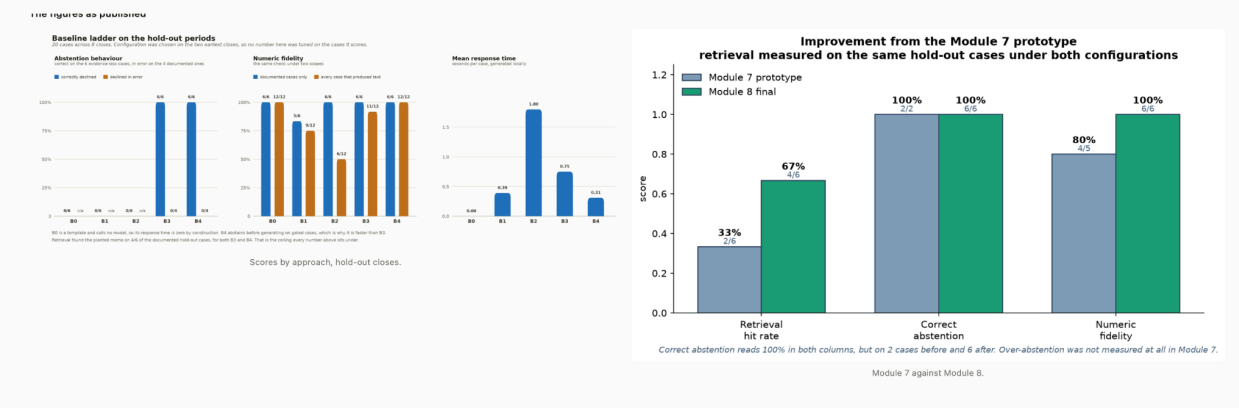
What moved retrieval

Configuration	Hits	Cases	Rate	Cases still missed
Module 7 baseline	2	6	0.333	P2402-1, P2412-1, P2508-1, P2510-1
Module 8 principled	4	6	0.667	P2508-1, P2510-1
k=4 variant (tuned)	5	6	0.833	P2508-1

Source results/retrieval_config_comparison.csv , written 28 August 2026. The gain comes from excluding prior-period commentary from the causes axis, which is a category correction rather than a tuning pass. The k=4 variant is reported separately and labelled tuned, because its only justification was a case whose failure had already been inspected.

Improvement over the Module 7 prototype

Area	Module 7 prototype	Module 8 final	Evidence
Retrieval hit rate (same hold-out cas	2/6 (33%)	4/6 (67%)	results/retrieval_config_comparison.csv; precedent documents excluded from the causes axis
Numeric fidelity	4/5 (C3 derived a threshold)	6/6	materiality thresholds now supplied as verified figures, so the model has no reason to compute one
Correct abstention	2/2	6/6	evidence-less stratum grown from 2 cases to 8, of which 6 fall in the hold-out periods reported here
Over-abstention (abstained with evi	not measured	0/4	metric added; measuring only correct abstention had hidden a gate that abstained on everything
Test cases	8 hand-picked	20 across 8 consecutive close periods	data/test_cases.csv
Comparison arms	1 (B3 only)	5 (B0 to B4)	results/benchmark_results.csv
Validation design	none, configured and reported on the same cases	temporal split, configured on 2 periods and reported on 8	an absolute threshold scored 4/4 on the config periods and then failed on 100% of hold-out cases
Corpus	12 documents	15 documents	D16 FX, D17 revolver and covenant, D18 stale launch brief added
Materiality rules	dollar thresholds only	dollar thresholds plus the covenant trigger	D01 4.3; a \$104K interest movement is immaterial in dollars yet reportable on covenant headroom



Every answer the system produced

The full text of every generated answer, with the score it received. This is the evidence behind the table at the top of the page: the numbers there are counts over these rows.

Approach

B0 Fixed template, no model B1 Model with no evidence B2 Model with the whole document set B3 Model with searched evidence **B4 Model with searched evidence and an evidence check**

Case	Close	Case type	Account	Declined	Figures traced	Seconds	Answer
P2402-1	2024-02	version sensitive	4000 - Revenue	no	yes	1.38	Headline: \$900,000 (+6.0%) variance due to favourable price effect. Cause: su
P2402-2	2024-02	immaterial	6200 - Marketing Operating Expense	no	yes	0.00	[no commentary produced; immaterial]
P2402-3	2024-02	evidence less	5000 - Cost of Goods Sold	yes	yes	1.41	Headline: \$745,000 (+7.9%) variance in Cost of Goods Sold. Cause: unsupport
P2405-1	2024-05	documented	6500 - R&D Operating Expense	no	yes	1.06	Headline: \$281,000 (+41.6%) variance in R&D Operating Expense [n] Cause: su
P2405-2	2024-05	immaterial	6200 - Marketing Operating Expense	no	yes	0.00	[no commentary produced; immaterial]
P2405-3	2024-05	evidence less	6100 - Legal & Professional	yes	yes	0.00	6100 - Legal & Professional variance \$371,000 (+58.6%) in 2024-05 [1]. Cause:
P2409-1	2024-09	evidence less	6100 - Legal & Professional	yes	yes	0.00	6100 - Legal & Professional variance \$2,400,000 (+379.1%) in 2024-09 [1]. Cau